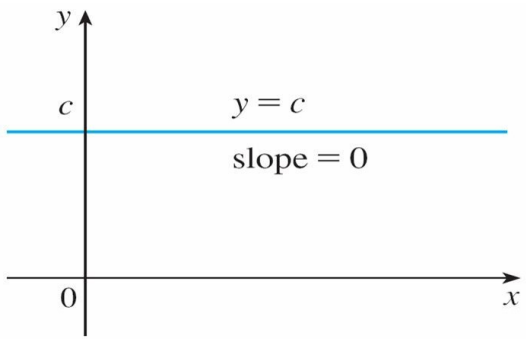
CHAPTER 3 DIFFERENTIAL RULES

3.1 Derivatives of Polynomials and Exponential Functions

- **Rules for Finding Derivatives**

1	• $\dfrac{d}{dx}c = 0$
2	• $\dfrac{d}{dx}x = 1$
3	• $\dfrac{d}{dx}x^n = nx^{n-1}$, n is a positive integer.

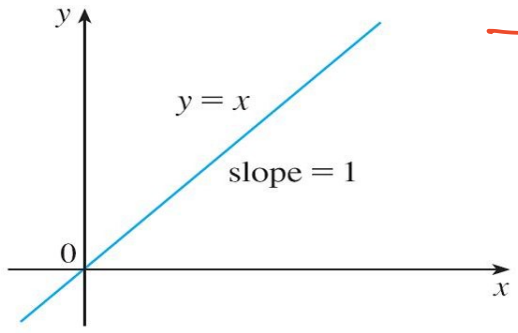


① $f(x) = C$

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0}\frac{C-C}{h} = \lim_{h\to 0} 0$$

$$= 0 \ \#$$



② $f(x) = x$

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0}\frac{x+h-x}{h}$$

$$= \lim_{h\to 0}\frac{h}{h} = \lim_{h\to 0} 1 = 1 \ \#$$

- For any real number n, $\dfrac{d}{dx}x^n = nx^{n-1}$.

1 • $D_x[kf(x)] = kD_x f(x)$

2 • $D_x[f(x) \pm g(x)] = D_x f(x) \pm D_x g(x)$

① let

$$\frac{d}{dx}[k f(x)] = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \quad *$$

$$= \lim_{h\to 0} \frac{k f(x+h) - k f(x)}{h} = \lim_{h\to 0} \frac{k[f(x+h) - f(x)]}{h}$$

$$= k \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= k \frac{d}{dx}[f(x)] \quad **$$

DIY

Example 3.1. Find the derivatives of the following functions:

(a) $y = 5x^2 + 7x - 6$

(b) $y = 4x^6 - 3x^5 - 10x^2 + 5x + 16$

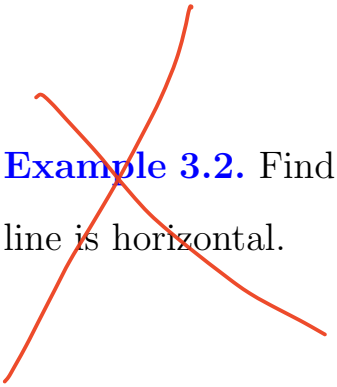
$y' = 24x^5 - 15x^4 - 20x + 5$

(c) $y = \dfrac{x^2+1}{x} = x + \dfrac{1}{x} = x + x^{-1}$

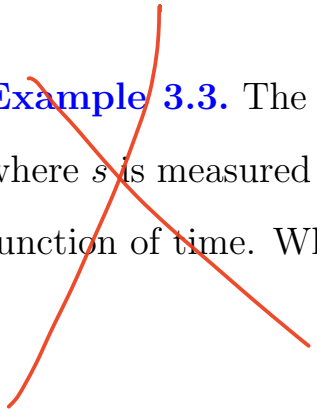
$\dfrac{dy}{dx} = 1 - x^{-2} = 1 - \dfrac{1}{x^2}$

(d) $y = \dfrac{x+1}{\sqrt{x}} = \dfrac{x}{\sqrt{x}} + \dfrac{1}{\sqrt{x}} = x^{\frac{1}{2}} + x^{-\frac{1}{2}}$

$\dfrac{dy}{dx} = \dfrac{1}{2}x^{-\frac{1}{2}} - \dfrac{1}{2}x^{-\frac{3}{2}} = \dfrac{1}{2\sqrt{x}} - \dfrac{1}{2\sqrt{x^3}}$



Example 3.2. Find the points on the curve $y = x^4 - 6x^2 + 4$ where the tangent line is horizontal.



Example 3.3. The equation of motion of a particle is $s = 2t^3 - 5t^2 + 3t + 4$, where s is measured in centimeters and t in seconds. Find the acceleration as a function of time. What is the acceleration after 2 seconds?

- **Derivatives of Exponential Functions**

- **The Definition of e**

e is the number such that $\lim_{h\to 0} \dfrac{e^h - 1}{h} = 1.$

- **Derivative of Natural Exponential Function**

$$\frac{d}{dx}e^x = e^x \quad (x)' = e^x$$

Example 3.4. At what point on the curve $y = e^x$ is the tangent line parallel to the line $y = 2x$?

Exercises Page 180 3, 5, 7, 11, 13, 17, 19, 21, 23, 25, 29, 31, 33, 35, 37, 49, 55, 57, 59, 65, 75.

3.2 Product and Quotient Rules

Example 3.5. Given $g(x) = x$, $h(x) = x^2$, and $f(x) = g(x)h(x)$, find $D_x g(x)$, $D_x h(x)$, $D_x f(x)$, and show that $D_x[g(x)h(x)] \neq [D_x g(x)]\,[D_x h(x)]$.

$g(x) = x$, $h(x) = x^2$, $f(x) = g(x)h(x)$

$g'(x) = 1$, $h'(x) = 2x$, $f(x) = x^3$, $f'(x) = 3x^2$

$D_x[g(x)h(x)] \neq [D_x g(x)][D_x h(x)]$

$3x^2 \neq 1\,(2x) = 2x$

- **Product Rule**

> If u and v are differentiable functions of x,
>
> $$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx} = uv' + vu'$$

DIY ① ②

Example 3.6. Find the derivative of $y = (3x^2-5)(2x^4-x)$ by using the product rule. Check the answer by first expanding the product and then take derivative.

Example 3.7. Given $f(x) = xe^x$,

(a) Find $f'(x)$.

(b) Find $f^{(n)}(x)$.

$f'(x) = x(e^x)' + e^x(x)' = xe^x + e^x = e^x(x+1)$

$f''(x) = e^x(1) + (x+1)e^x = e^x(x+2)$

$f'''(x) = e^x(1) + (x+2)e^x = e^x(x+3)$

$\vdots$

$f^{(n)}(x) = e^x(x+n)$

$(\sqrt{t})' = \frac{1}{2\sqrt{t}}$

Example 3.8. Given $f(t) = \sqrt{t}(a + bt)$, find $f'(t)$.

$f(t) = t^{\frac{1}{2}}(a+bt)$

$f'(t) = t^{\frac{1}{2}}(b) + \frac{1}{2}t^{-\frac{1}{2}}(a+bt)$

$= b\sqrt{t} + \frac{a+bt}{2\sqrt{t}}$

Example 3.9. Given $f(x) = \sqrt{x}g(x)$, where $g(4) = 2$, $g'(4) = 3$, find $f'(4)$.

$f'(4) = ?$

$f'(x) = \sqrt{x}\ g'(x) + \frac{1}{2\sqrt{x}}g(x)$

$f'(4) = \sqrt{4}\ g'(4) + \frac{1}{2\sqrt{4}}g(4)$

$= 2(3) + \frac{1}{2(2)}(2) = 6 + \frac{1}{2} = \frac{13}{2}$

- **Quotient Rule**

If u and v are differentiable functions of x,

$$\frac{d}{dx}\frac{u}{v} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2} = \frac{vu' - uv'}{v^2}$$

at the point where $v(x) \neq 0$.

Example 3.10. Find $\frac{d}{dx}\left(\frac{3x-5}{x^2+7}\right)$.

$$\frac{d}{dx}\left(\frac{3x-5}{x^2+7}\right) = \frac{(x^2+7)(3x-5)' - (3x-5)(x^2+7)'}{(x^2+7)^2}$$

$$= \frac{(x^2+7)(3) - (3x-5)(2x)}{(x^2+7)^2}$$

$$= \frac{3x^2+21-6x^2+10x}{(x^2+7)^2}$$

$$= \frac{-3x^2+10x+21}{(x^2+7)^2}$$

Example 3.11. Find the equation of the tangent line to the curve $y = \dfrac{e^x}{1+x^2}$ at the point $\left(1, \dfrac{e}{2}\right)$.

Exercises Page 188 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 31, 43, 45, 47, 49, 55.

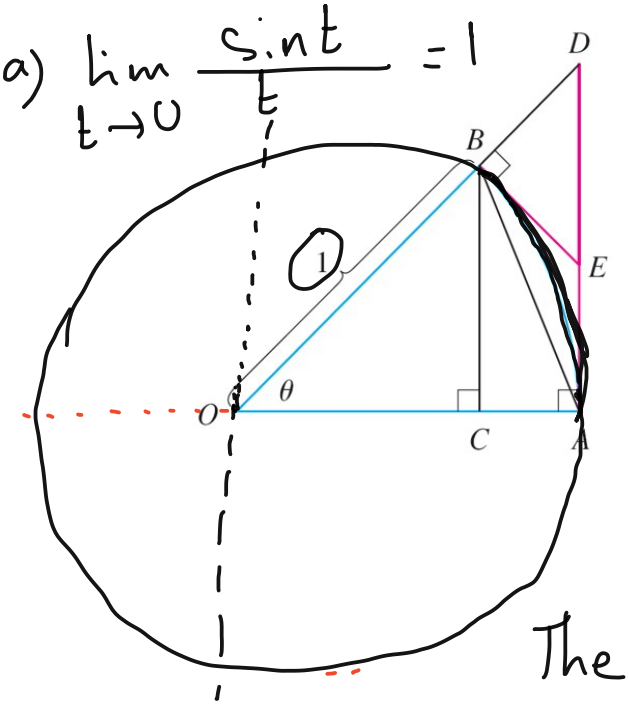
3.3 Derivatives of Trigonometric Functions

- **Special Trigonometric Limits**

(a) $\lim_{t\to 0} \dfrac{\sin t}{t} = 1$ § $\lim_{t\to 0} \dfrac{t}{\sin t} = \lim_{t\to 0} \dfrac{1}{\frac{\sin t}{t}} = \dfrac{1}{1} = 1$

(b) $\lim_{t\to 0} \dfrac{1-\cos t}{t} = 0.$ § $\lim_{t\to 0} \dfrac{t}{1-\cos t} = \lim_{t\to 0} \dfrac{\frac{1}{t}}{\frac{1-\cos t}{t}} = \dfrac{1}{0} = \pm\infty$

(a) $\lim_{t\to 0} \dfrac{\sin t}{t} = 1$



$\hat{OBC}$, $\sin\theta = \dfrac{BC}{OB} = \dfrac{BC}{1} = BC$

$\hat{ODA}$, $\tan\theta = \dfrac{AD}{OA} = \dfrac{AD}{1} = AD$

Arc length, $\hat{AB} = r\theta = \theta$

The geometric fact is that,

$$|BC| < \hat{AB} < |AD|$$

$$\sin\theta < \theta < \tan\theta$$

$$\sin\theta < \theta < \frac{\sin\theta}{\cos\theta}$$

$$1 < \frac{\theta}{\sin\theta} < \frac{1}{\cos\theta}$$

$$\cos\theta < \frac{\sin\theta}{\theta} < 1$$

$\lim_{\theta \to 0} \cos\theta = 1$ & $\lim_{\theta \to 0} 1 = 1$

By squeeze theorem, $\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1$ ✻

(b) $\lim_{t \to 0} \frac{1 - \cos t}{t} = 0$

$\frac{0}{0}$

$$L.H.S = \lim_{t \to 0} \frac{1 - \cos t}{t} \times \frac{1 + \cos t}{1 + \cos t} = \lim_{t \to 0} \frac{1 - \cos^2 t}{t(1 + \cos t)}$$

$$= \lim_{t \to 0} \frac{\sin^2 t}{t(1 + \cos t)} = \lim_{t \to 0} \frac{\sin t}{t} \times \lim_{t \to 0} \frac{\sin t}{1 + \cos t}$$

(1) (0)

$$= 0$$ ✻

Example 3.12. Find $\lim_{x\to 0} \dfrac{\sin 3x}{x}$.

$$\lim_{x\to 0} \frac{\sin 3x}{x} \times \frac{3}{3} = \lim_{x\to 0} \frac{3\sin 3x}{3x}$$

$x \to 0$, $3x \to 0$

$$= 3 \lim_{3x\to 0} \frac{\sin 3x}{3x} = 3$$

Example 3.13. Find $\lim_{t\to 0} \dfrac{1-\cos t}{\sin t}$.

$$\lim_{t\to 0} \frac{1-\cos t}{\sin t} \times \frac{t}{t} = \lim_{t\to 0} \frac{1-\cos t}{t} \times \frac{t}{\sin t}$$

$$= \lim_{t\to 0} \frac{1-\cos t}{t} \times \lim_{t\to 0} \frac{t}{\sin t} = 0 \times 1 = 0$$

Example 3.14. Find $\lim_{x\to 0} \dfrac{\sin 4x}{\tan x}$.

$$\lim_{x\to 0} \frac{\sin 4x}{\tan x} = \lim_{x\to 0} \frac{\sin 4x}{\tan x} \times \frac{4x}{4x} = \lim_{4x\to 0} \frac{\sin 4x}{4x} \times \lim_{x\to 0} \frac{4x}{\tan x}$$

$\frac{0}{0}$

$$= \lim_{x\to 0} \frac{4x}{\tan x} = \lim_{x\to 0} \frac{4x}{\frac{\sin x}{\cos x}} = \lim_{x\to 0} \frac{4x\cos x}{\sin x}$$

$$= 4 \lim_{x\to 0} \frac{x}{\sin x} \times \lim_{x\to 0} \cos x = 4$$

- **Theorem**

  - $\dfrac{d}{dx}\sin x = \cos x$ $(x)' = \cos x$
  - $\dfrac{d}{dx}\cos x = -\sin x$ $(x)'$

Example 3.15. Find $\dfrac{d}{dx}(3\sin x - 2\cos x)$.

$\dfrac{d}{dx}(3\sin x - 2\cos x) = 3\cos x + 2\sin x$

Example 3.16. Find the equation of the tangent line to the curve $y = 3\sin x$ at the point $(\pi, 0)$.

Example 3.17. Find $\dfrac{d}{dx}\left(x^2 \sin x\right)$.

$$\frac{d}{dx}(x^2 \sin x) = x^2(\cos x) + 2x \sin x$$

$$= x(2\sin x + x\cos x)$$

Example 3.18. Find $\dfrac{d}{dx}\left(\dfrac{1+\sin x}{\cos x}\right)$.

$$\frac{d}{dx}\left(\frac{1+\sin x}{\cos x}\right)$$

① ②

$$\frac{d}{dx}\left(\frac{1+\sin x}{\cos x}\right) = \frac{\cos x(\cos x) - (1+\sin x)(-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin x + \sin^2 x}{\cos^2 x} = \frac{1+\sin x}{\cos^2 x}$$

$$= \sec^2 x + \tan x \sec x$$

$$\frac{d}{dx}(\sec x + \tan x)$$

$$= \sec x \tan x + \sec^2 x$$

- **Theorem**

  - $\frac{d}{dx}\tan x = \sec^2 x$ $(x)'$
  - $\frac{d}{dx}\cot x = -\csc^2 x$ $(x)'$
  - $\frac{d}{dx}\sec x = \sec x \tan x$ $(x)'$
  - $\frac{d}{dx}\csc x = -\csc x \cot x$ $(x)'$

Example 3.19. Find the equation of the tangent line to the curve $y = \tan x$ at the point $\left(\frac{\pi}{4}, 1\right)$.

Example 3.20. Find $\frac{d}{dx}\left(x^n \cot x\right)$ where n is an integer.

$$\frac{d}{dx}\left(x^n \cot x\right) = x^n\left(-\csc^2 x\right) + n x^{n-1} \cot x$$

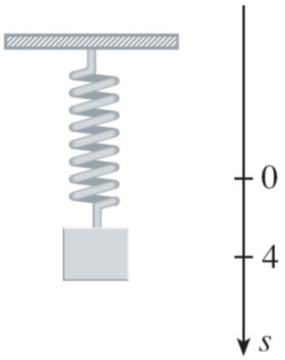
$$= x^{n-1}\left(n \cot x - x \csc^2 x\right)$$

Example 3.21. Find all the points on the graph of $y = \sin^2 x$ where the tangent line is horizontal.

Example 3.22. An object at the end of a vertical spring is stretched 4 cm beyond its rest position and released at time $t = 0$. Its position at time t is

$$s = f(t) = 4\cos t$$

Find the velocity and acceleration of the object at time t and analyze the motion of the object.



Exercises Page 196 1, 3, 5, 7, 9, 11, 13, 15, 17, 18, 19, 20, 23, 33, 35, 39, 41, 43, 45, 51, 57 (*Hint*: $s = r\theta$, draw the bisector of the angle θ and find $\sin(\frac{\theta}{2})$).

3.4 The Chain Rule

Example 3.23. Find $\frac{d}{dx}(x^2+2x)^3$.

(1) $y = (x^2+2x)^3$

$y = (x^2+2x)(x^2+2x)(x^2+2x)$

$y = x^6 + 6x^5 + 12x^4 + 8x^3$

$\frac{dy}{dx} = 6x^5 + 30x^4 + 48x^3 + 24x^2$

(2) $y = (x^2+2x)^3$

let $u = x^2+2x$, $y = u^3$

$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

$= 3u^2(2x+2)$

$= 3(x^2+2x)^2(2x+2)$

- **Chain Rule**

Let $y = f(u)$ and $u = g(x)$. If g is differentiable at x and f is differentiable at $u = g(x)$, then the composite function $f \circ g$, defined by $(f \circ g)(x) = f(g(x))$, is differentiable at x and

$$(f \circ g)'(x) = f'(g(x))\, g'(x)$$

or

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Example 3.24. If $y = \left(2x^2 - 4x + 1\right)^{60}$, find $D_x y$.

$$D_x y = 60\left(2x^2 - 4x + 1\right)^{59}(4x - 4)$$

Example 3.25. If $y = \dfrac{1}{\sqrt[3]{2x^5 - 7}}$, find $\dfrac{dy}{dx}$.

$$y = (2x^5 - 7)^{-\frac{1}{3}}$$

$$\frac{dy}{dx} = -\frac{1}{3}(2x^5 - 7)^{-\frac{4}{3}}(10x^4) = \frac{-10x^4}{3(2x^5 - 7)^{\frac{4}{3}}}$$

Example 3.26. Find $D_t\left(\dfrac{t^3 - 2t + 1}{t^4 + 3}\right)^{13}$.

$$D_t\left(\frac{t^3 - 2t + 1}{t^4 + 3}\right)^{13} = 13\left(\frac{t^3 - 2t + 1}{t^4 + 3}\right)^{12}\left[\frac{(t^4 + 3)(3t^2 - 2) - (t^3 - 2t + 1)(4t^3)}{(t^4 + 3)^2}\right]$$

$$= 13\left(\frac{t^3 - 2t + 1}{t^4 + 3}\right)\left[\frac{-t^6 + 6t^4 - 4t^3 + 9t^2 - 6}{(t^4 + 3)^2}\right]$$

Example 3.27. If $y = \sin 2x$, find $\dfrac{dy}{dx}$.

$\dfrac{dy}{dx} = \cos 2x\,(2) = 2\cos 2x$

Example 3.28. Find $F'(y)$ if $F(y) = y\sin(y^2)$. $uv' + vu'$

$F'(y) = y\left(\cos y^2\,(2y)\right) + \sin y^2\,(1)$

$= 2y^2\cos y^2 + \sin y^2$

Example 3.29. Find $D_x\dfrac{x^2(1-x)^3}{1+x}$.

Example 3.30. Find $\frac{d}{dx}e^{\sin x}$.

$$\frac{d}{dx}e^{\sin x} = e^{\sin x}\left[\cos x\right](1) = \cos x\, e^{\sin x}$$

Example 3.31. Find $\frac{d}{dx}a^x$, where $a > 0$, $a \neq 1$.

$$y = a^x = e^{\ln a^x} = e^{x \ln a}$$

$$\frac{dy}{dx} = e^{x \ln a}(\ln a) = e^{\ln a^x}\ln a = \boxed{a^x \ln a}$$

Example 3.32. Find $\frac{d}{d\theta}2^{\sec\theta}$.

$$\frac{d}{d\theta}\left(2^{\sec\theta}\right) = 2^{\sec\theta}(\sec\theta\tan\theta)\ln 2$$

Example 3.33. If F is differentiable, find

(a) $D_x\left[F(x^3)\right]$

(b) $D_x\left[(F(x))^3\right]$.

(a) $D_x\left[F(x^3)\right] = F'(x^3)\left[3x^2\right]$

(b) $D_x\left[F(x)\right]^3 = 3\left[F(x)\right]^2 F'(x)$

Example 3.34. Find $D_x \sin^3(4x)$.

$D_x\left[\sin(4x)\right]^3 = 3\sin^2(4x)\cos(4x)(4)$

$= 12\sin^2(4x)\cos(4x)$

HW **Example 3.35.** Find $D_x \sin\left[\cos\left(x^2\right)\right]$. $= -2x\sin x^2 \cos\left[\cos(x^2)\right]$

Exercises Page 204 7, 9, 11, 13, 15, 17, 25, 27, 29, 31, 43, 51, 53, 61, 71, 73, 79, 81.

3.5 Implicit Differentiation

- The relation $y^3+7y=x^3$ can define y as a function of x. It is called an **implicit function**.

Example 3.36. Find $\dfrac{dy}{dx}$ if $y^3+7y=x^3$.

$3y^2y' + 7y' = 3x^2$

$y'(3y^2+7) = 3x^2$

$\dfrac{dy}{dx} = \dfrac{3x^2}{3y^2+7}$

Example 3.37. Find the equation of the tangent line to the curve $4x^2y - 3y = x^3$ at the point $(1, 1)$.

Example 3.38. At what point in the first quadrant is the tangent line to the curve $x^3 + y^3 = 6xy$ horizontal?

Example 3.39. Given $\sin(x+y) = y^2 \cos x$, find $\frac{dy}{dx}$.

H.W

$$= \frac{y^2 \sin x + \cos(x+y)}{2y\cos x - \cos(x+y)}$$

Example 3.40. Given $x^4 + y^4 = 16$, find y''.

$$4x^3 + 4y^3 y' = 0$$

$$4y^3 y' = -4x^3$$

$$y' = -\frac{x^3}{y^3}$$

$$y'' = \frac{y^3(-3x^2) + x^3(3y^2 y')}{y^6}$$

$$y'' = \frac{-3x^2y^3 + 3x^3y^2\left(-\frac{x^3}{y^3}\right)}{y^6} = \frac{-3x^2y^3 - 3\frac{x^6}{y}}{y^6}$$

$$y'' = \frac{\dfrac{-3x^2y^4 - 3x^6}{y}}{y^6} = \frac{-3x^2y^4 - 3x^6}{y^7} \quad *$$

$$= \frac{-3x^2(y^4 + x^4)}{y^7}$$

- **Derivatives of Inverse Trigonometric Functions**

- $D_x \sin^{-1} x = \dfrac{1}{\sqrt{1-x^2}}$ → 1 → derivative of the angle; $\sqrt{1-x^2}$ → (angle)^2
- $D_x \cos^{-1} x = -\dfrac{1}{\sqrt{1-x^2}}$
- $D_x \tan^{-1} x = \dfrac{1}{1+x^2}$

Example 3.41. Differentiate $y = \dfrac{1}{\sin^{-1} x}$.

$= (\sin^{-1} x)^{-1}$

$\dfrac{dy}{dx} = -(\sin^{-1} x)^{-2} \left(\dfrac{1}{\sqrt{1-x^2}} \right)$

$= \dfrac{-1}{(\sin^{-1} x)^2 \sqrt{1-x^2}}$

$x^{\frac{1}{2}}$

Example 3.42. Differentiate $y = x \arctan \sqrt{x}$.

$= x \tan^{-1} \sqrt{x}$

$y' = x \dfrac{\frac{1}{2\sqrt{x}}}{1+x} + \tan^{-1} \sqrt{x}$ (1)

$= \dfrac{\sqrt{x}}{2(1+x)} + \tan^{-1} \sqrt{x}$

Exercises Page 215 5, 7, 9, 11, 13, 15, 17, 19, 27, 35, 49, 51, 57, 63, 75.

3.6 Derivatives of Logarithmic Functions

- **Derivatives of Logarithmic Functions**

• $D_x \ln x = \dfrac{1}{x}$
• $D_x \log_a x = \dfrac{1}{x \ln a}$

Example 3.43. Find $D_x \ln \sqrt{x}$.

$$D_x \ln \sqrt{x} = \frac{\frac{1}{2\sqrt{x}}}{\sqrt{x}} = \frac{1}{2\sqrt{x}\sqrt{x}} = \frac{1}{2x}$$

Example 3.44. Find $D_x \ln\left(x^2 - x - 2\right)$.

$$D_x \ln(x^2 - x - 2) = \frac{2x - 1}{x^2 - x - 2}$$

Example 3.45. Show that

$$D_x \ln|x| = \frac{1}{x}$$

(1) When $x > 0$

$$D_x \ln|x| = D_x \ln x = \frac{1}{x}$$

(2) When $x < 0$

$$D_x \ln|x| = D_x \ln(-x) = \frac{-1}{-x} = \frac{1}{x}$$

Example 3.46. Find $\frac{d}{dx}\ln(1+\cos x)$.

$$\frac{d}{dx}\ln(1+\cos x) = \frac{-\sin x}{1+\cos x}$$

Example 3.47. Find $\frac{d}{dx}\sqrt{\ln x}$.

$$\frac{d}{dx}(\ln x)^{\frac{1}{2}} = \frac{1}{2}(\ln x)^{-\frac{1}{2}}\left[\frac{1}{x}\right] = \frac{1}{2x\sqrt{\ln x}}$$

$$\ln x^a = a\ln x$$
$$\ln\frac{a}{b} = \ln a - \ln b$$

Example 3.48. Find $\frac{dy}{dx}$ if $y = \ln\sqrt[3]{\frac{x-1}{x^2}}$, $x > 1$.

$$y = \ln\left(\frac{x-1}{x^2}\right)^{\frac{1}{3}} = \frac{1}{3}\ln\frac{x-1}{x^2} = \frac{1}{3}\left[\ln(x-1) - \ln x^2\right]$$

$$= \frac{1}{3}\left[\ln(x-1) - 2\ln x\right]$$

$$\frac{dy}{dx} = \frac{1}{3}\left[\frac{1}{x-1} - \frac{2}{x}\right]$$

Example 3.49. Differentiate $y = \frac{\sqrt{1-x^2}}{(x+1)^{\frac{2}{3}}}$.

$$y = \frac{\sqrt{1-x^2}}{(x+1)^{\frac{2}{3}}}$$

$$\ln y = \ln\frac{\sqrt{1-x^2}}{(x+1)^{\frac{2}{3}}} = \ln(1-x^2)^{\frac{1}{2}} - \ln(x+1)^{\frac{2}{3}}$$

$$\ln y = \frac{1}{2}\ln(1-x^2) - \frac{2}{3}\ln(x+1)$$

$$\frac{y'}{y} = \frac{1}{2}\,\frac{-2x}{1-x^2} - \frac{2}{3}\,\frac{1}{x+1}$$

$$y' = y\left[\frac{-x}{1-x^2} - \frac{2}{3x+3}\right]$$

$$y' = \frac{\sqrt{1-x^2}}{(x+1)^{\frac{2}{3}}}\left[\frac{-x}{1-x^2} - \frac{2}{3x+3}\right]$$

Example 3.50. Differentiate $y = x^x$.

$$y = x^x = e^{\ln x^x} = e^{x \ln x}$$

$$y' = e^{x \ln x}\left[x \frac{1}{x} + \ln x\,(1)\right]$$

$$y' = x^x\left[1 + \ln x\right]$$

Example 3.51. Differentiate $y = \left(x^2+1\right)^{\sin x}$.

$$\ln y = \ln\left(x^2+1\right)^{\sin x}$$

$$\ln y = \sin x \ln\left(x^2+1\right)$$

$$\frac{y'}{y} = \sin x \frac{2x}{x^2+1} + \ln\left(x^2+1\right)\cos x$$

$$y' = y\left[\frac{2x \sin x}{x^2+1} + \cos x \ln\left(x^2+1\right)\right]$$

$$y' = \left(x^2+1\right)^{\sin x}\left[\frac{2x \sin x}{x^2+1} + \cos x \ln\left(x^2+1\right)\right]$$

- ***e* as a Limit**

$$\lim_{x\to 0}(1+x)^{\frac{1}{x}} = e$$

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n} = e$$

Example 3.52. Find $\lim_{n\to\infty}\left(\frac{n+2}{n}\right)^{n}$.

Example 3.53. Find $\lim_{n\to\infty}\left(\frac{n-2}{n}\right)^{n}$.

Exercises Page 223 3, 5, 7, 9, 13, 15, 19, 33, 39, 41, 47, 56.

3.7 Rates of Change

Example 3.54. The position of a particle is given by

$$s = f(t) = t^3 - 6t^2 + 9t$$

where s is measured in meters and $t \geq 0$ is measured in seconds.

(a) Find the velocity at time t.

(b) What is the velocity after 2 s? After 4 s?

(c) When is the particle at rest?

(d) When is the particle moving to the right?

(e) Find the total distance travelled by the particle in the first 5 seconds.

(f) Find the acceleration at time t.

(g) When is the particle speeding up? When is it slowing down?

Exercises Page 233 1(a)(b)(c)(d)(e)(g)(i).

3.9 Related Rates

- If a variable y depends on time t, then its derivative $\dfrac{dy}{dt}$ is called a time rate of change.

Example 3.55. Air is pumped into a spherical balloon so that its volume increases at a rate of 100 cm^3/s. How fast is the radius of the balloon increasing when the diameter is 50 cm?

Example 3.56. A small balloon is released at a point 150 m away from an observer, who is on level ground. If the balloon goes straight up at a rate of 8 m/s, how fast is the distance from the observer to the balloon increasing when the balloon is 50 m high?

Example 3.57. Water is pouring into a conical tank at the rate of 8 m^3 per second. If the height of the tank is 12 m and the radius of its circular opening is 6 m, how fast is the water level rising when the water is 4 m deep?

Example 3.58. An airplane flying north at 320 km per hour passes a certain town at noon. A second airplane going east at 300 km per hour is directly over the same town 15 minutes later. If the airplanes are flying at the same altitude, how fast will they be separating at 1:15 P.M.?

Example 3.59. A woman standing on a cliff is watching a motorboat through a telescope as the boat approaches the shoreline directly below her. If the telescope is 100 m above the water level and the boat is approaching at 8 m per second, at what rate is the angle of the telescope changing when the boat is 100 m from the shore?

Example 3.60. As the sun sets behind a 40-meter building, the building's shadow grows. How fast is the shadow growing when the sun's rays make an angle of 45°?

Exercises Page 249 3, 5, 17, 22 (1.01 m/s), 25, 32 (−0.25 rad/s), 45.

3.10 Linear Approximations and Differentials

- $$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

  When $x - a$ is small, $f'(a) \approx \dfrac{f(x) - f(a)}{x - a}$, and yields

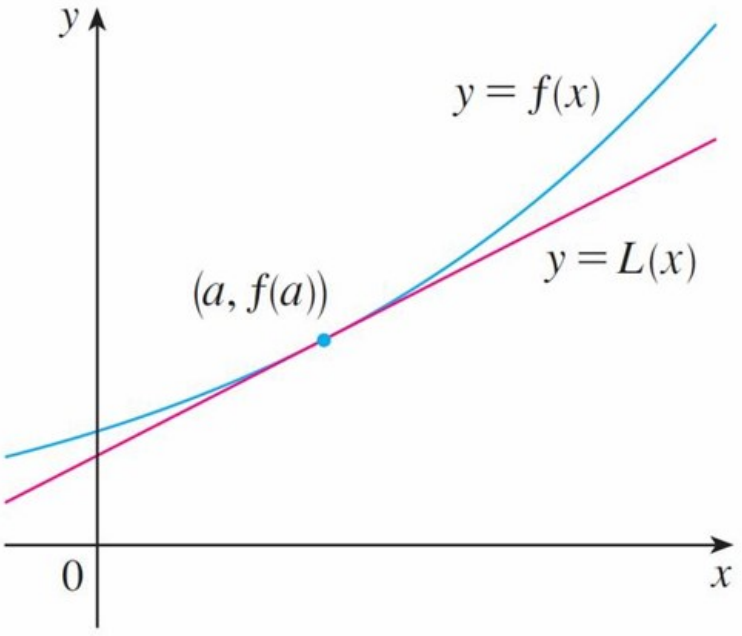
  $$f(x) - f(a) \approx (x - a) f'(a)$$

  $$f(x) \approx f(a) + f'(a)(x - a)$$

- **Linearization**

$$L(x) = f(a) + f'(a)(x - a)$$

is called the **linearization** of f at a.



Example 3.61. Find approximations for $\sqrt{4.6}$ and $\sqrt{8.2}$.

- **Differentials**

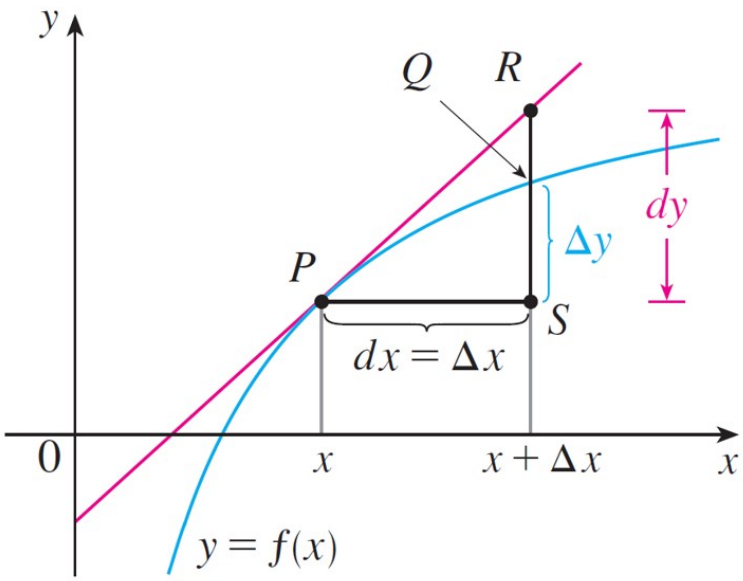
Let $y = f(x)$ be a differentiable function of the independent variable x.

- Δx is an arbitrary increment in the independent variable x.
- dx, called the **differential** of the independent variable x, is equal to Δx.
- Δy is the actual change in the variable y as x changes from x to $x + \Delta x$; that is

$$\Delta y = f(x + \Delta x) - f(x)$$

- dy, called the **differential** of the dependent variable y, is defined as

$$dy = f'(x)dx$$



Example 3.62. Compute the value of Δy and dy if $f(x) = x^3 + x^2 - 2x + 1$ and x changes from 2 to 2.05.

Example 3.63. Approximate the increase in the area of a soap bubble when its radius increases from 3 cm to 3.025 cm.

Example 3.64. The side of a cube is measured as 11.4 cm with a possible error of ± 0.05 cm. Evaluate the volume of the cube and give an estimate for the possible error in this value.

Example 3.65. Poiseuille's Law for blood flow says that the volume flowing through an artery is proportional to the fourth power of the radius, that is, $V = kR^4$. By how much must the radius be increased in order to increase the blood flow by 50%?

Exercises Page 256 19, 25, 33, 35, 40 (20%).

3.11 Hyperbolic Functions

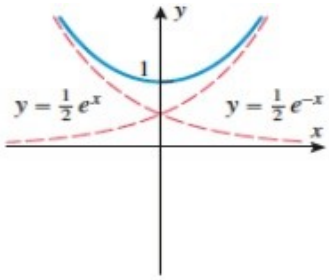
- **Definition**

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

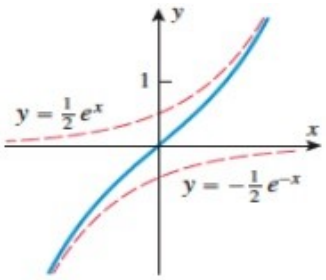
$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\coth x = \frac{\cosh x}{\sinh x}$$



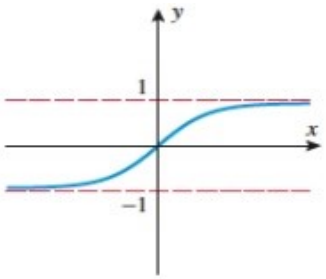
$y = \cosh x$

(a)



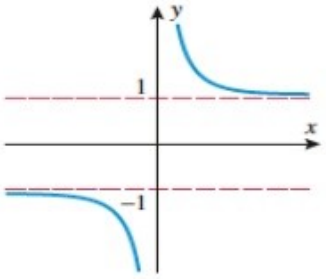
$y = \sinh x$

(b)



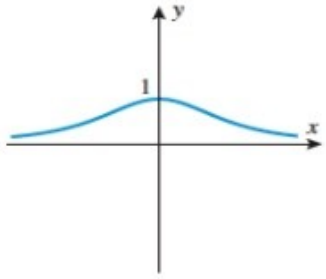
$y = \tanh x$

(c)



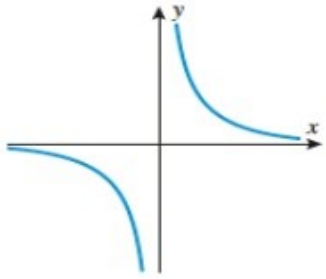
$y = \coth x$

(d)



$y = \operatorname{sech} x$

(e)



$y = \operatorname{csch} x$

(f)

- **Identity of Hyperbolic Functions**

$$\cosh^2 x - \sinh^2 x = 1$$

- **Derivatives of Hyperbolic Functions**

  - $D_x \sinh x = \cosh x$
  - $D_x \cosh x = \sinh x$

Example 3.66. Find $D_x \cosh^2(3x - 1)$.

Exercises Page 264 24(a), 31 $\left(\dfrac{1}{2\sqrt{x}\cosh^2\sqrt{x}}\right)$, 33, 35, 37.